Yale Quan

Yale.Quan@usm.edu | yalequan.github.io | Google Scholar

Faculty Appointments

University of Southern Mississippi, Hattiesburg, MS 2026–present
Assistant Professor of Education (tenure track)
Research, Evaluation, Statistics, and Assessment
School of Education, College of Education and Human Sciences

Education

University of Washington, Seattle, WA
Ph.D. in Measurement and Statistics 2026
Dissertation: *Culturally responsive measurement: Machine learning and artificial intelligence approaches to DIF detection, parameter estimation, and scoring*
Advisor: Chun Wang

California State University, Long Beach, Long Beach, CA
M.S. in Applied Statistics 2020
B.S. in Criminal Justice 2013

Peer-Reviewed Journal Articles

5. **Quan, Y.**, & Wang, C. (2026). Transitive differential item functioning clustering: A graph-theoretic approach to identifying many-group partial measurement invariance. *Psychological Methods*. Advance online publication. https://doi.org/10.1037/met0000873

4. Wang, C., **Quan, Y.**, & Arthur, D. (2026). Review of cognitive diagnostic models (CDMs): Recent methodological advancements for addressing practical challenges. *British Journal of Mathematical and Statistical Psychology*. Advance online publication. https://doi.org/10.1111/bmsp.70066

3. **Quan, Y.**, & Wang, C. (2026). Using multilabel classification neural network to detect intersectional DIF with small sample sizes. *British Journal of Mathematical and Statistical Psychology*. Advance online publication. https://doi.org/10.1111/bmsp.70041

2. **Quan, Y.**, & Wang, C. (2026). Calibrating multidimensional assessments with structural missingness: An application of a multiple-group higher-order IRT model. *Applied Psychological Measurement*, *50*(4–5), 155–172. https://doi.org/10.1177/01466216251415011

1. **Quan, Y.**, & Wang, C. (2025). Collapsing or not? A practical guide to handling sparse responses for polytomous items. *Methodology*, *21*(1), 46–73. https://doi.org/10.5964/meth.14303

Peer-Reviewed Conference Proceedings

1. Parker, M. C., Ciou, S. Y., **Quan, Y.**, Ren, H., Wang, C., & Li, M. (2026). Investigating answer choice bias within a college-level introductory computing assessment. In *Proceedings of the 57th ACM Technical Symposium on Computer Science Education V.1* (pp. 817–823). Association for Computing Machinery. https://doi.org/10.1145/3770762.3772622

Manuscripts and Conference Papers Under Review

3. **Quan, Y.**, & Wang, C. *Long-InterDIFNet: A long-format neural network for transportable intersectional DIF flagging* [Manuscript under review]. Journal of Educational and Behavioral Statistics.

2. Parker, M. C., **Quan, Y.**, Wang, C., & Li, M. *Can AI write the test? The irreplaceable role of humans in computing assessment creation* [Conference paper under review]. ACM CHI Conference on Human Factors in Computing Systems.

1. Shao, K., Wilson, D., **Quan, Y.**, & Makhsous, S. *Designing needs- and attention-aware AI chatbots for engineering education: Insights from psychological outcomes* [Conference paper under review]. ACM CHI Conference on Human Factors in Computing Systems.

Preprints

1. **Quan, Y.**, & Ren, H. (2025). *Sample size and assessment length recommendations for the diagnostic facet status model.* PsyArXiv. https://doi.org/10.31234/osf.io/zr3je_v1

Software and Code

Grants, Fellowships, and Awards

AI Pro Tool Access Program Mini-Grant 2026
University Libraries, University of Southern Mississippi
Role: Principal Investigator
Supports research on using large language models to develop psychological survey items.

Three-Year Community Partner Fellowship Award 2022–2025
University of Washington, Seattle, WA

Invited Talks and Workshops

3. **Quan, Y.** (2026, April 24). *A gentle introduction to developing neural networks in Python* [Invited workshop]. Center for Social Science Computation and Research, University of Washington, Seattle, WA, United States.

2. **Quan, Y.** (2025, October 23). *Machine learning methods for supervised single-label and multi-label classification* [Invited workshop]. Center for Social Science Computation and Research, University of Washington, Seattle, WA, United States.

1. **Quan, Y.** (2024, October 18). *Item response theory (IRT) model selection and applications* [Invited talk]. Behavioral Research Center for HIV, University of Washington, Seattle, WA, United States.

Peer-Reviewed Conference Presentations

9. **Quan, Y.**, & Wang, C. (2026, April 8–11). *Beyond DIF detection: A downstream clustering framework for small sample parameter estimation* [Paper presentation]. National Council on Measurement in Education Annual Meeting, Los Angeles, CA, United States.

8. **Quan, Y.**, & Wang, C. (2026, April 8–11). *Detecting small sample Q-matrix misspecification in the diagnostic facet status model* [Paper presentation]. National Council on Measurement in Education Annual Meeting, Los Angeles, CA, United States.

7. **Quan, Y.**, & Wang, C. (2025, July 15–18). *A neural network approach to small sample intersectional DIF detection* [Paper presentation]. International Meeting of the Psychometric Society, Minneapolis, MN, United States.

6. **Quan, Y.**, Ren, H., & Wang, C. (2025, March 29–30). *Constructing a machine learning model for binary predictions with incomplete, imbalanced data and non-linear effects* [Paper presentation]. A Meeting of Methodologies, online.

5. **Quan, Y.**, & Sager, M. (2025, February 17–19). *Identifying core formal assessment competencies and informal learning outcomes of data science and data literacy* [Paper presentation]. Data Science Education K–12: Research to Practice Conference, San Antonio, TX, United States.

4. **Quan, Y.**, & Wang, C. (2024, November 17–18). *A new item fit test for the diagnostic facet status model (DFSM)* [Paper presentation]. Annual Pacific Northwest Research on Psychometrics and Applied Statistics Conference, Pullman, WA, United States.

3. **Quan, Y.**, & Wang, C. (2024, April 11–14). *Parameter recovery from higher order item response theory models with structural missingness* [Paper presentation]. National Council on Measurement in Education Annual Meeting, Philadelphia, PA, United States.

2. **Quan, Y.**, & Wang, C. (2024, April 11–14). *Collapsing or not? A practical guide to handling sparse responses for polytomous items* [Paper presentation]. American Educational Research Association Annual Meeting, Philadelphia, PA, United States. https://doi.org/10.3102/2102869

1. **Quan, Y.**, & Wang, C. (2023, April 12–15). *The effects of sample size and collapse direction on parameter recovery* [Poster presentation]. National Council on Measurement in Education Annual Meeting, Chicago, IL, United States.

University Research Talks and Seminars

5. **Quan, Y.** (2025, May 9). *Using categorical structural equation models to identify facets of student belonging* [Research presentation]. Community Partner Doctoral Fellowship Research Presentation, University of Washington, Seattle, WA, United States.

4. **Quan, Y.**, & Ren, H. (2024, December 4). *Sample size and assessment length recommendations for the diagnostic facet status model* [Research presentation]. Center for Statistics and the Social Sciences Student Research Presentation, University of Washington, Seattle, WA, United States.

3. **Quan, Y.** (2024, June 4). *Introduction to Bayesian item response theory* [Research presentation]. Center for Statistics and the Social Sciences Student Research Presentation, University of Washington, Seattle, WA, United States.

2. **Quan, Y.**, & Xiao, T. (2024, March 11). *The effects of measurement error on multilevel linear growth model parameter estimates* [Research presentation]. Center for Statistics and the Social Sciences Student Research Presentation, University of Washington, Seattle, WA, United States.

1. **Quan, Y.** (2021, December 3). *Clustering education data using K-medoids with partitioning around the medoids algorithm* [Seminar presentation]. Measurement and Statistics Seminar, University of Washington, Seattle, WA, United States.

Research Experience

Project HARLI, Seattle, WA 2026
Quantitative Researcher and Interviewer

National Council of Teachers of Mathematics, Seattle, WA 2026
Quantitative Researcher

Center for Social Science Computation and Research, Seattle, WA 2025–2026
Statistical Research Consultant, University of Washington (also 2021)

Commented [CV1]: Add one or two factual contribution statements for each role as the living CV develops. Prioritize the research question, your methods and responsibilities, and a resulting paper, report, instrument, or other product. Project HARLI and NCTM currently have only role titles. For Equal Opportunity Schools, connect the student-belonging presentation to the fellowship if that relationship is accurate. Retain project location separately from employer location if needed. Add current Southern Miss research projects when you can describe their scope.

Equal Opportunity Schools, Seattle, WA 2022–2025
Community Partner Doctoral Fellow

Data Science 4 Everyone, Chicago, IL 2024
Graduate Research Assistant

Teaching Experience

University of Southern Mississippi 2026–present
Measurement in Educational Research
Quantitative Research Methods II

University of Washington 2021–2022
Teaching Assistant, College of Education
Basic Educational Statistics
Teaching Assistant, Department of Psychology
Fundamentals of Psychological Research

California State University, Long Beach 2017–2021
Lecturer, Department of Mathematics and Statistics 2020–2021
Statistics for Everyday Life
Teaching Associate, Department of Mathematics and Statistics 2018–2020
Statistics for Everyday Life
Essential Algebra A/B
The Power of Mathematics
Supplemental Instructor, Department of Mathematics and Statistics 2017–2018
Business Calculus
Calculus I

Commented [CV2]: For the living CV, add course numbers, undergraduate or graduate level, terms taught, and instructor-of-record responsibilities where applicable. The current draft retains the roles and courses supplied in the source. Clarify the overlapping 2020–2021 Lecturer and 2018–2021 Teaching Associate appointments. Add advising or mentoring as a separate section if you have relevant activities to record.

Mentoring

Service

University Service

Journal Reviewer

Psychometrika 2026–present
Educational and Psychological Measurement 2026–present
British Journal of Mathematical and Statistical Psychology 2026–present
Journal of Educational and Behavioral Statistics 2026–present
Journal of Educational Measurement 2025–present

Conference Proposal Reviewer

American Educational Research Association 2023–present
National Council on Measurement in Education 2022–present

Professional Training and Certifications

University of Pennsylvania, Philadelphia, PA 2025
Data Science Methods for Digital Learning Platforms Certification

American Institutes for Research, Arlington, VA 2023
National Assessment of Educational Progress Winter Data Training

American Institutes for Research, Arlington, VA 2021
National Assessment of Educational Progress Process Data Training Series

Professional Memberships

National Council on Measurement in Education 2022–present
American Educational Research Association, Division D 2022–present